Scattering Theory

D - 8.4.26

Q2. Given $V(r) = \lambda\,\delta(r-a)$

$$E = \frac{\hbar^2 k^2}{2m}$$

Radial Schrodinger equation:

$$-\frac{\hbar^2}{2m}\frac{d^2u(r)}{dr^2} + V(r)\,u(r) = E\,u(r)$$

$$\Rightarrow \quad -\frac{\hbar^2}{2m}u''(r) + \lambda\,\delta(r-a)\,u(r) = \frac{\hbar^2 k^2}{2m}u(r)$$

$$u''(r) + k^2 u(r) = \frac{2m\lambda}{\hbar^2}\,\delta(r-a)\,u(r)$$

$$= \xi\,\delta(r-a)\,u(r) \quad \left(\xi = \frac{2m\lambda}{\hbar^2}\right) \qquad \text{(i)}$$

In $r<a$:

here the delta function is zero. We need to solve

$$u''(r) + k^2 u(r) = 0$$

$$\Rightarrow u_1(r) = A\sin(kr) + B\cos(kr)$$

Since the physical radial wave function $R(r) = \frac{u(r)}{r}$ diverges near origin if $B \neq 0$, we get

$$u_1(r) = A\sin(kr)$$

In $r>a$: The delta function is also zero here. The solution can be written in a general form

$$u_2(r) = A'\sin(kr) + B'\cos(kr)$$

Let's define two new constants, an overall amplitude C and phase shift angle δ_0:

$$A' = C\cos\delta_0, \quad B' = C\sin(\delta_0)$$

$\therefore \quad u_>(r) = \sin(kr + \delta_0)$

<u>Boundary condition:</u>

$$u_<(a) = u_>(a)$$

$$\Rightarrow \quad A \sin(ka) = C \sin(ka + \delta_0) \quad \text{——— (ii)}$$

Integrating eq (i),

$$\lim_{\epsilon \to 0} \int_{a-\epsilon}^{a+\epsilon} u''(r)\,dr + \lim_{\epsilon \to 0} k^2 \int_{a-\epsilon}^{a+\epsilon} u(r)\,dr = \xi \lim_{\epsilon \to 0} \int_{a-\epsilon}^{a+\epsilon} \delta(r-a)\, u(r)\,dr$$

$$u_>'(a) - u_<'(a) + 0 = \xi\, u(a)$$

$$Ck \cos(ka + \delta_0) - Ak \cos(ka) = \xi A \sin(ka) \quad \text{——— (iii)}$$

replacing C,

$$kA\cot(ka + \delta_0)\sin(ka) - kA\cos(ka) = \xi A \sin(ka)$$

$$k\cot(ka + \delta_0) - k\cot(ka) = \xi$$

<u>Solving δ_0:</u>

$$\tan \delta_0 = \frac{-\xi \sin^2(ka)}{k + \xi \sin(ka)\cos(ka)}$$

b) S-wave cross section $\sigma_0 = \frac{4\pi}{k^2}\sin^2(\delta_0)$

using $\sin^2(\delta_0) = \dfrac{\tan^2(\delta_0)}{1 + \tan^2(\delta_0)}$

$$\sigma_0 = \frac{4\pi}{k^2}\left[\frac{\xi^2 \sin^4(ka)}{k^2 + 2k\xi \sin(ka)\cos(ka) + \xi^2 \sin^2(ka)}\right]$$

③ In low energy limit ($ka \ll 1$), $\sin(ka) \simeq ka$, $\cos(ka) \simeq 1$

$$\therefore \quad \sigma_0 \simeq \frac{4\pi}{k^2} \frac{\xi^2 (ka)^4}{k^2 + 2k^2 \xi a + \xi^2 k^2 a^2}$$

$$= 4\pi a^2 \frac{\xi^2 a^2}{(1+\xi a)^2} = 4\pi a^2 \frac{1}{(1+1/\xi a)^2}$$

For $\lambda \to \infty$, $\xi \to \infty$. $\Rightarrow$ $\sigma_0 \simeq 4\pi a^2$ (hard sphere limit)

c) As $\lambda \to \infty$, $\xi \to \infty$.

$$\tan \delta_0 = \frac{-\sin^2 ka}{k/\xi + \sin ka \cos ka} \simeq -\cot(ka) = -ka$$

Which is exactly the s-wave phase shift of a hard sphere of radius a. The delta shell becomes impenetrable in this limit.

Q3. Scattering amplitude $f(\theta,\phi) = -\frac{m}{2\pi\hbar^2} \int e^{-i\vec{k}_f \cdot \vec{r}} V(\vec{r}) \psi(\vec{r}) d^3r$

First Born approximation: This assumes the scattering potential $V(r)$ is weak enough and the approximation of the full wave function $\psi(\vec{r})$ with the incident plane wave $e^{i\vec{k}_i \cdot \vec{r}}$ (essentially unperturbed)

$$\Rightarrow f^{(1)}(\theta,\phi) = -\frac{m}{2\pi\hbar^2} \int e^{i(\vec{k}_i - \vec{k}_f)\cdot\vec{r}} V(\vec{r}) \, d^3r$$

(4) Let, the momentum transfer $\vec{q} = \vec{k}_i - \vec{k}_f$. Assuming elastic scattering ($|\vec{k}_i| = |\vec{k}_f| = k$) $|\vec{q}| = 2k \sin\frac{\theta}{2}$

For spherically symmetric potential

$$f^{(1)}(\theta) = -\frac{m}{2\pi\hbar^2} \int_0^\infty dr\, r^2 \int_{-1}^{1} d(\cos\theta)\, e^{iqr\cos\theta}\, V(r) \int_0^{2\pi} d\varphi$$

$$= -\frac{m}{\hbar^2} \int_0^\infty dr\, r^2\, V(r)\, \frac{e^{iqr} - e^{-iqr}}{iqr}$$

$$= -\frac{2m}{q\hbar^2} \int_0^\infty dr\, r\, V(r) \sin(qr)$$

(a) $V(r) = V_0 e^{-r^2/\sigma^2}$

using $\int_0^\infty r \sin(qr)\, e^{-r^2/\sigma^2}\, dr = -\frac{\sqrt{\pi}\, q\, \sigma^3}{4} e^{-q^2\sigma^2/4}$

$$\therefore f^{(1)}(\theta) = -\frac{2m}{q\hbar^2} V_0 \left(-\frac{\sqrt{\pi}\, q\, \sigma^3}{4} e^{-q^2\sigma^2/4}\right)$$

$$= \frac{m V_0 \sqrt{\pi}\, \sigma^3}{2\hbar^2} e^{-q^2\sigma^2/4}$$

$$= \frac{m V_0 \sqrt{\pi}\, \sigma^3}{2\hbar^2} e^{-k^2\sigma^2 \sin^2\theta/2}$$

b) Differential cross-section

$$\frac{d\sigma}{d\Omega} = |f^{(1)}(\theta)|^2 = \left(\frac{m V_0 \sqrt{\pi}\, \sigma^3}{2\hbar^2}\right)^2 e^{-2k^2\sigma^2 \sin^2\theta/2}$$

This explicitly depends on V_0^2, hence independent of the sign of V_0 at this order.

(c) Total cross section $\sigma^{(1)}_{tot} = \int |f^{(1)}(\theta)|^2\, d\Omega$

$$= 2\pi \int_0^\pi |f^{(1)}(\theta)|^2 \sin\theta\, d\theta$$

5)

$$= 2\pi \left(\frac{m V_0 \sqrt{\pi}\, \sigma^3}{2\hbar^2} \right)^2 \int_0^{\pi} \sin\theta \; e^{-2k^2\sigma^2 \sin^2\theta/2} \, d\theta$$

Now,

$$q = 2k \sin\frac{\theta}{2}$$

$$q^2 = 4k^2 \sin^2\frac{\theta}{2} = 2k^2(1-\cos\theta)$$

$$2q\,dq = 2k^2 \sin\theta \, d\theta$$

$$\sin\theta \, d\theta = \frac{q\,dq}{k^2}$$

$$\sigma^{(1)}_{tot} = 2\pi \left(\frac{m V_0 \sqrt{\pi}\, \sigma^3}{2\hbar^2} \right)^2 \int_0^{2k} \frac{q\,dq}{k^2} \, e^{-q^2\sigma^2/2}$$

$$= \frac{m^2 \pi^2 V_0^2 \sigma^6}{2\hbar^4 k^2} \frac{1}{\sigma^2} \left(1 - e^{-2k^2\sigma^2}\right)$$

$$= \frac{m^2 \pi^2 V_0^2 \sigma^4}{2\hbar^4 k^2} \left(1 - e^{-2k^2\sigma^2}\right)$$

For fixed σ and large k, $e^{-2k^2\sigma^2} \approx 0$.

So, $$\frac{1 - e^{-2k^2\sigma^2}}{k^2} \approx \frac{1}{k^2}$$

$\therefore$ Total cross-section decreases with increasing k.

Physically a higher k value corresponds to a faster-moving particle, it sees the potential as a slowly varying perturbation. The probability of the particle being scattered is reduced, leading to a smaller cross-section at higher energies.

S wave scattering $(l=0)$:

(6) Q4. The radial Schrodinger equation for $u(r) = rR(r)$ is

(a)

$$-\frac{\hbar^2}{2m}\frac{d^2u(r)}{dr^2} + V(r)u(r) = E\, u(r)$$

Inside the well $(r<a)$: $V(r) = -V_0$

$$\frac{d^2u_<}{dr^2} + K^2 u_< = 0 \quad , \text{ where } K^2 = \frac{2m(E+V_0)}{\hbar^2} = k^2 + \frac{2mV_0}{\hbar^2}$$

solution: $u_<(r) = A\sin(Kr)$

Outside the well $(r>a)$: $V(r) = 0$

$$\frac{d^2u_>}{dr^2} + k^2 u_> = 0 \quad , \quad k^2 = \frac{2mE}{\hbar^2}$$

solution: $u_>(r) = C\sin(kr+\delta_0)$

Boundary condition:

$$u_<(a) = u_>(a) \Rightarrow A\sin(Ka) = C\sin(ka+\delta_0)$$

$$u_<'(a) = u_>'(a) \Rightarrow AK\cos(Ka) = Ck\cos(ka+\delta_0)$$

$$\Rightarrow \quad K\cot(Ka) = k\cot(ka+\delta_0)$$

$$\tan(ka+\delta_0) = \frac{k}{K}\tan(Ka)$$

Solving for $\tan\delta_0$:

$$\tan\delta_0 = \frac{k\tan(Ka) - K\tan(ka)}{K + k\tan(Ka)\tan(ka)}$$

(b) Scattering length $a_s = -\lim_{k\to 0} \frac{\tan\delta_0}{k}$

As $k \to 0$, $\tan ka \sim ka$, $K \to \sqrt{\frac{2mV_0}{\hbar^2}} = K_0$

$$\lim_{k\to 0} \frac{\tan\delta_0}{k} = \frac{\tan(K_0 a)}{K_0} - a$$

$$\therefore a_s = a\left(1 - \frac{\tan(K_0 a)}{K_0 a}\right)$$

when $\tan(K_0 a) = K_0 a$, $a_s = 0$

True zero resonance condition:

For the scattering length to physically diverge, $\tan(K_0 a)$ must approach infinity.

$$\therefore K_0 a = \frac{\pi}{2}, \frac{3\pi}{2}, \ldots$$

$$\tan\delta_0 = \frac{k - K\cot(Ka)\tan(ka)}{K\cot(Ka) + k\tan(ka)}$$

$$= \frac{k - 0}{k\tan(ka)} = \frac{1}{ka}$$

as $k \to 0$, $\tan\delta_0 \to \infty \Rightarrow \delta_0 = \frac{\pi}{2}$

The s-wave cross section $\sigma_0 = \frac{4\pi}{k^2}\sin^2(\delta) = \frac{4\pi}{k^2}$

(c) $K_0 a = \frac{\pi}{2}$ (condition for shallowest possible well at zero energy resonance)

$$\sqrt{\frac{2mV_0}{\hbar^2}}\, a = \frac{\pi}{2} \Rightarrow V_0 = \frac{\pi^2\hbar^2}{8a^2 m}$$

Scattering length $a_s = a - \frac{\tan(k_0 a)}{k_0} = a - \frac{\infty}{k_0} = -\infty$

Q5.

(a) Optical theorem,

$$\sigma_{total} = \frac{4\pi}{k} \operatorname{Im}[f(0)]$$

In the first born approximation,

$$f^{(1)}(\theta) = -\frac{2m}{\hbar^2 q} \int_0^\infty r \sin(qr)\, V(r)\, dr$$

For a real potential, $f^{(1)}(\theta)$ is real. Since $\sigma_{tot} > 0$, this violates the optical theorem.

(b) The second Born amplitude

$$f^{(2)}(k_f, k_i) = -\frac{m}{2\pi\hbar^2} \int \frac{d^3k'}{(2\pi)^3} \frac{\langle k_f | V | k' \rangle \langle k' | V | k_i \rangle}{E - \frac{\hbar^2 k'^2}{2m} + i\epsilon}$$

$k_f = k_i = k$

$$\langle k | V | k' \rangle = \tilde{V}(k - k')$$

For a real central potential, $\tilde{V}(q) = \tilde{V}^*(q)$

$$f^{(2)}(0) = -\frac{m}{2\pi\hbar^2} \int \frac{d^3k'}{(2\pi)^3} \frac{|\tilde{V}(\vec{k} - \vec{k}')|^2}{\frac{\hbar^2}{2m}(k^2 - k'^2) + i\epsilon}$$

$$\operatorname{Im}[f^{(2)}(0)] = -\frac{m}{2\pi\hbar^2} \int \frac{d^3k'}{(2\pi)^3} |\tilde{V}(\vec{k} - \vec{k}')|^2 \left[-\pi\, \delta\left(\frac{\hbar^2}{2m}(k^2 - k'^2)\right) \right]$$

$$= \frac{m^2}{2\hbar^4 k (2\pi)^3} \int k'^2\, dk'\, d\Omega'\, |\tilde{V}(\vec{k} - \vec{k}')|^2\, \delta(k - k')$$

$$= \frac{m^2 k}{8\pi^2 \hbar^4 \times 2\pi} \int d\Omega'\, |\tilde{V}(\vec{k} - \vec{k}')|^2$$

$$\frac{d\sigma^{(1)}}{d\Omega} = \left| -\frac{m}{2\pi\hbar^2}\tilde{V}(q)\right|^2 = \frac{m^2}{4\pi^2\hbar^4}\left|\tilde{V}(q)\right|^2$$

$$\int d\Omega' \, |\tilde{V}(q)|^2 = \frac{4\pi^2\hbar^4}{m^2}\sigma_{tot}^{(1)}$$

$$\therefore \; \mathrm{Im}\, f^{(2)}(0) = \frac{m^2 k}{8\pi^2\hbar^4 \times 2\pi}\frac{4\pi^2\hbar^4}{m^2}\sigma_{tot}^{(1)} = \frac{k}{4\pi}\sigma_{tot}^{(1)}$$

$\therefore$ Optical theorem is satisfied to second order in potential.

(c) $$V(r) = V_0 e^{-\mu r}/r$$

Fourier transform: $$\tilde{V}(q) = \frac{4\pi V_0}{q^2+\mu^2}$$

$$\therefore \; \frac{d\sigma^{(1)}}{d\Omega} = \left(\frac{2mV_0}{\hbar^2}\right)^2 \frac{1}{\left(4k^2\sin^2\frac{\theta}{2}+\mu^2\right)^2}$$

$$\sigma_{tot}^{(1)} = 2\pi\left(\frac{2mV_0}{\hbar^2}\right)^2 \int_0^{\pi} \frac{\sin\theta\, d\theta}{\left(4k^2\sin^2\frac{\theta}{2}+\mu^2\right)^2}$$

Let $u = \sin^2\frac{\theta}{2} \Rightarrow \sin\theta\, d\theta = 2\, du$

$$\sigma_{tot}^{(1)} = 4\pi\left(\frac{2mV_0}{\hbar^2}\right)^2 \int_0^1 \frac{du}{(4k^2u+\mu^2)^2}$$

$$= \frac{4\pi m^2 V_0^2}{\hbar^4 k^2}\left(\frac{1}{\mu^2} - \frac{1}{\mu^2+4k^2}\right)$$

$$= \frac{16\pi m^2 V_0^2}{\hbar^4\mu^2(\mu^2+4k^2)}$$

On the other hand, the imaginary part of $f^{(2)}(0)$ from part (b) equals $\frac{k}{4\pi}\sigma_{tot}^{(1)}$. A direct evaluation of $f^{(2)}(0)$ leads to the same result (check yourself).